

Tensile Strength Data

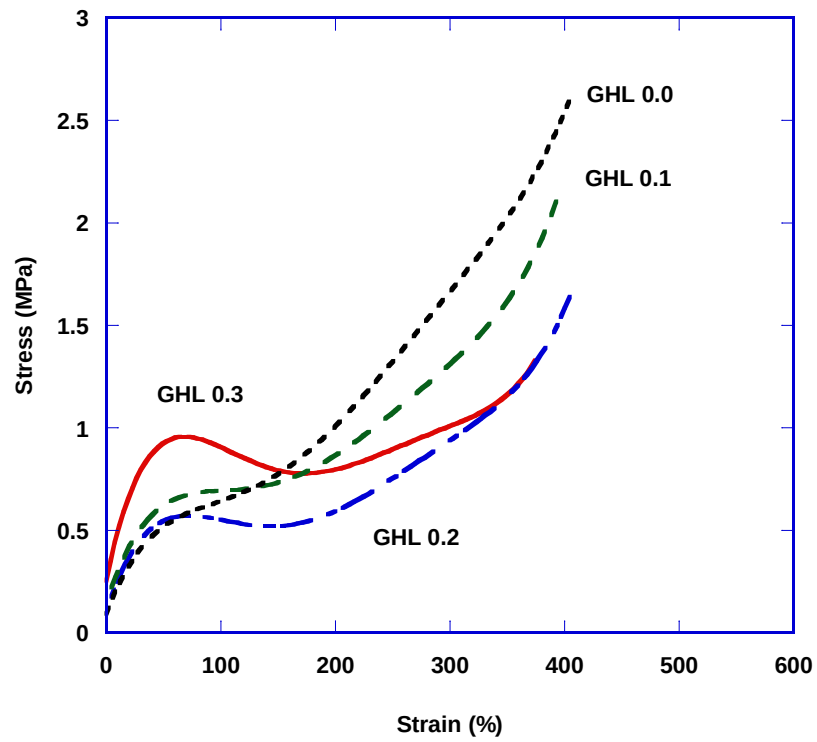


Figure 1: Engineering Stress for GHL

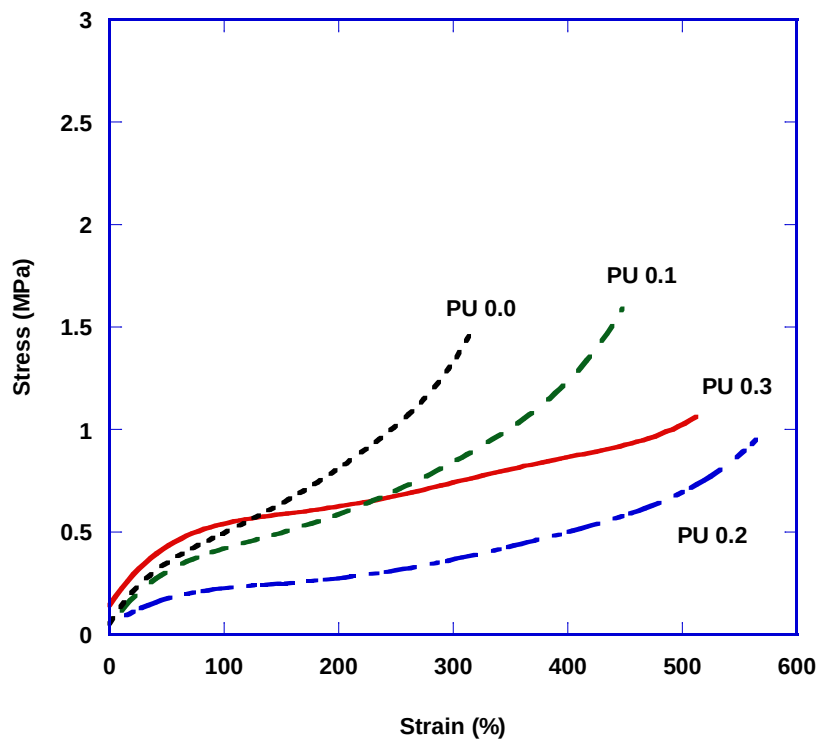


Figure 2: Engineering Stress for PU

Table 1: Summary of Engineering Stress Data

PU	Young's Modulus			
	Break Strength (MPa)	(MPa)	Toughness (MPa)	% Strain
0	1.57 ± 0.10	1.01 ± 0.19	2.31 ± 0.17	319 ± 12
0.1	1.49 ± 0.21	0.97 ± 0.11	2.97 ± 0.47	435 ± 34
0.2	0.88 ± 0.06	1.01 ± 0.13	2.16 ± 0.10	560 ± 19
0.3	1.16 ± 0.11	1.75 ± 0.11	3.85 ± 0.68	539 ± 69
GHL	Young's Modulus			
	Break Strength (MPa)	(MPa)	Toughness (MPa)	% Strain
0	2.56 ± 0.17	1.79 ± 0.10	4.61 ± 0.39	396 ± 18
0.1	2.10 ± 0.17	2.50 ± 0.20	3.98 ± 0.34	396 ± 14
0.2	1.61 ± 0.06	2.50 ± 0.32	3.08 ± 0.13	401 ± 10
0.3	1.21 ± 0.10	4.39 ± 0.27	3.08 ± 0.24	364 ± 13

Mooney Rivlin Model

Equation 1: Mooney Rivlin Equation

$$\sigma = \left(2C_1 + \frac{2C_2}{\lambda} \right) \left(\lambda - \frac{1}{\lambda^2} \right)$$

Equation 2: Relationship to cross link density

$$2C_1 = \frac{\rho RT}{M_c}$$

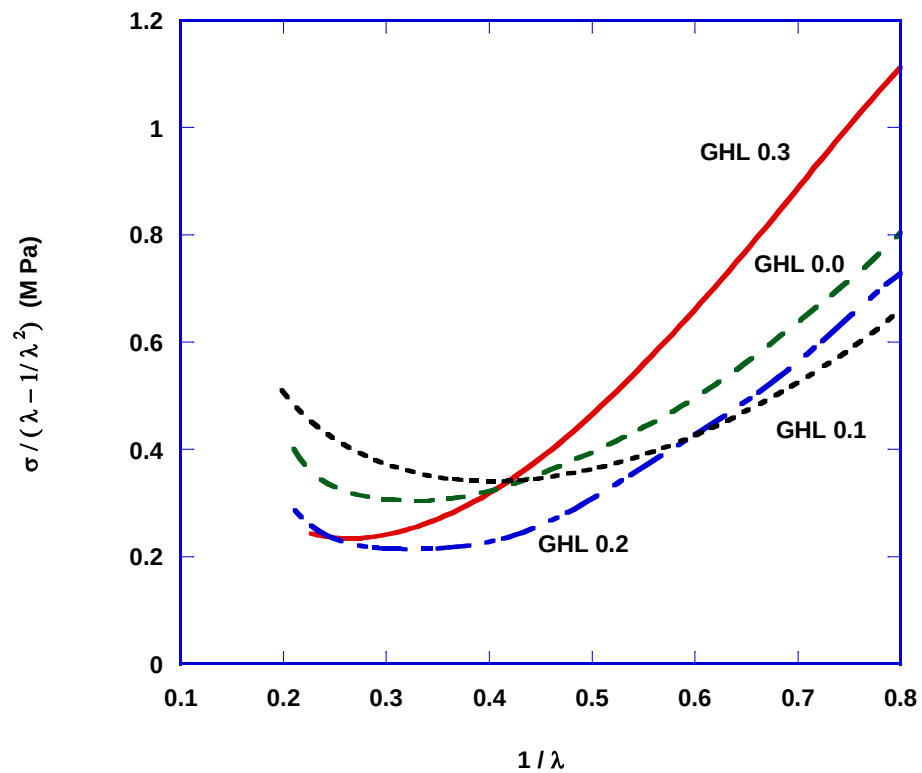


Figure 3: Mooney-Rivlin plot for GHl

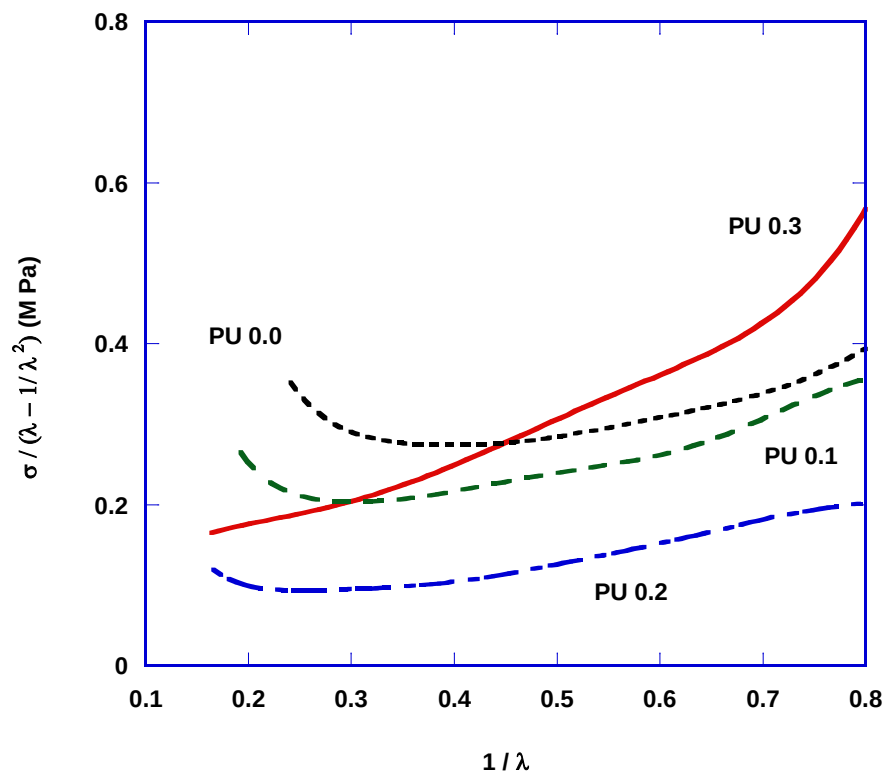


Figure 4: Mooney-Rivlin plot for PU

Table 2: Cross-linking density estimated from Mooney-Rivlin plots

Mc (kg / mol)		Mc (kg / mol)	
PU 0.0	5.9	GHL 0.0	4.8
PU 0.1	9.3	GHL 0.1	8.2
PU 0.2	16.1	GHL 0.2	8.1
PU 0.3	n/a	GHL 0.3	n/a

DSC DATA

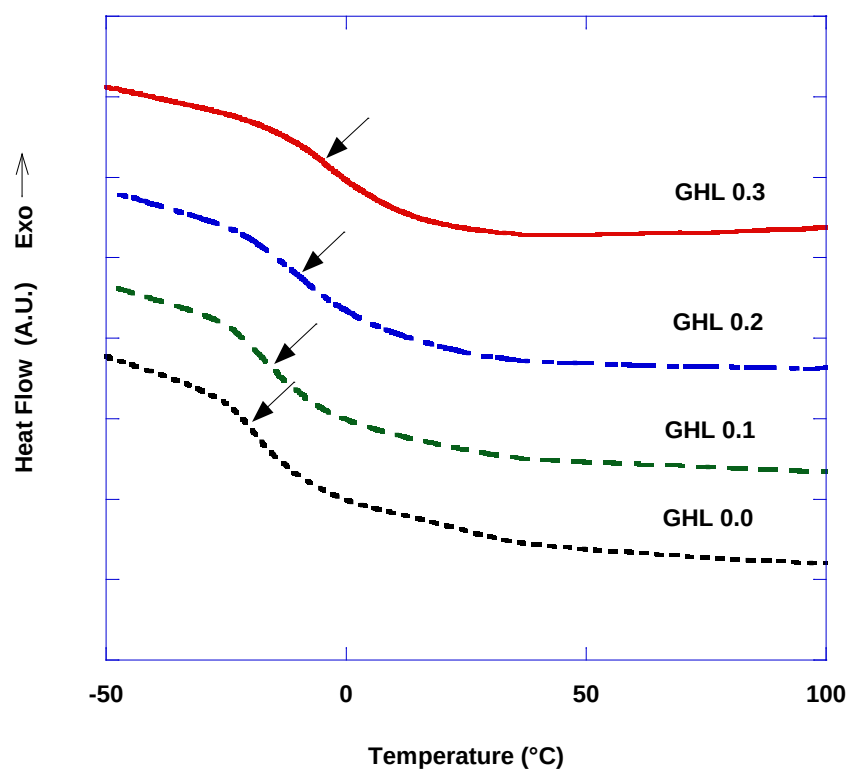


Figure 5: DSC thermograms of GHL films

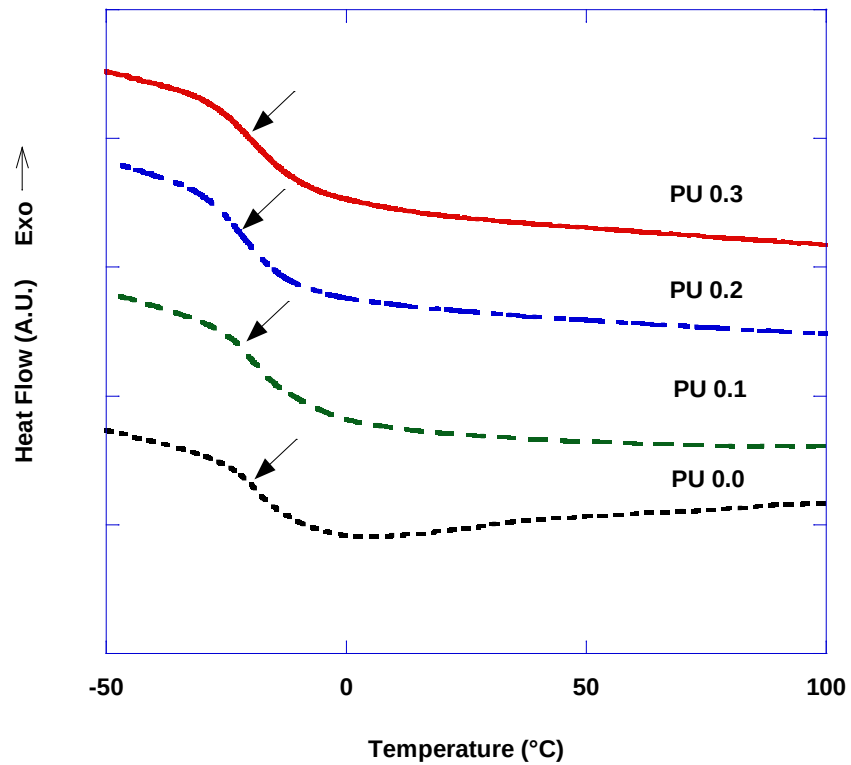


Figure 6: DSC thermograms of PU films

Table 3: Glass transition temperatures of GHL and PU films determined by DSC

PU	T _g (°C)	GHL	T _g (°C)
0.00	-18.60	0.00	-19.80
0.10	-20.80	0.10	-16.10
0.20	-21.70	0.20	-9.30
0.30	-19.60	0.30	-4.80

Table 4: Glass transition temperatures of GHL films (without HEA) and with different vinyl monomer ratios

HEA	ST	BA	T _g (°C)
0.00	25	75	-19.2
0.00	50	50	-19.8
0.00	75	25	-20

Table 5: Glass transition temperatures of GHL films (with HEA) and with different vinyl monomer ratios

HEA	ST	BA	Tg (°C)
0.30	25	75	-11.2
0.30	50	50	-4.8
0.30	75	25	-10.3

TGA DATA

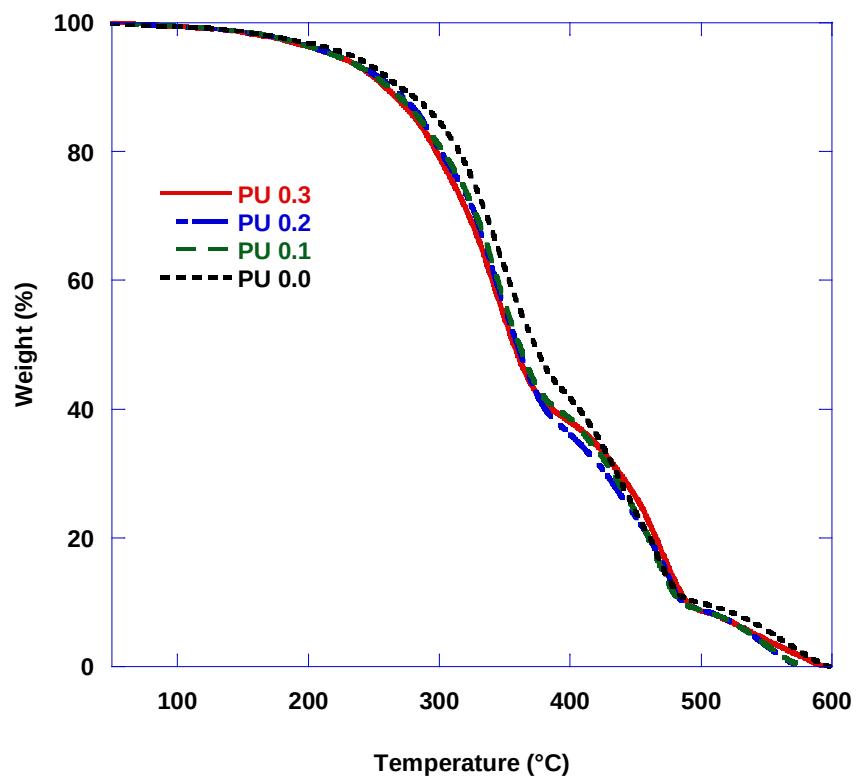


Figure 7: TGA curves for PU films with different amounts of HEA

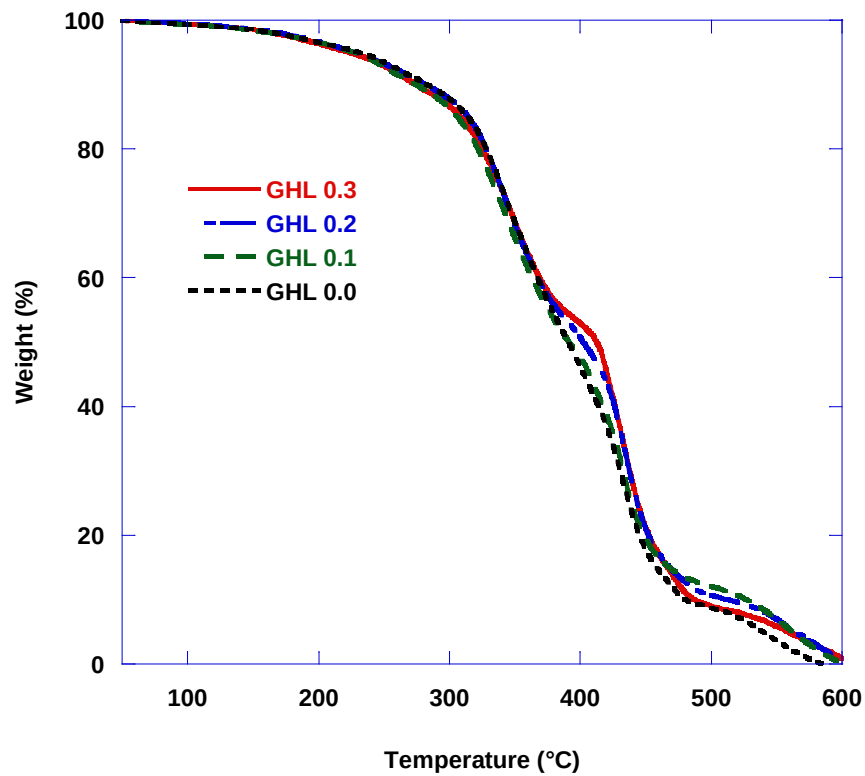


Figure 8: TGA curves for GHF films with different amounts of HEA

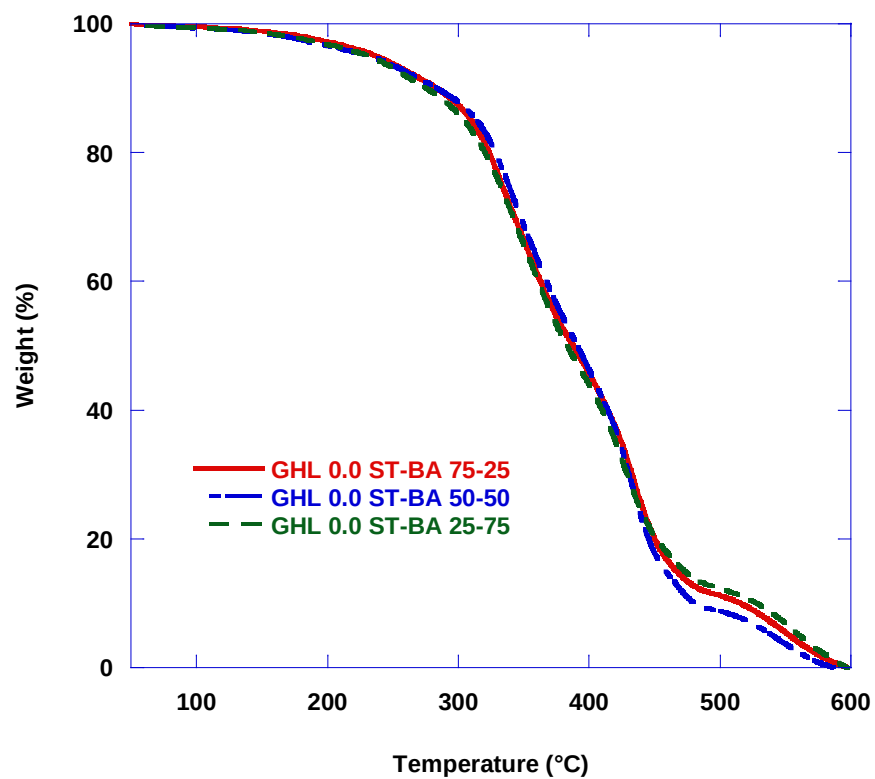


Figure 9: TGA curves for GHF films with different ratios of vinyl monomers (without HEA)

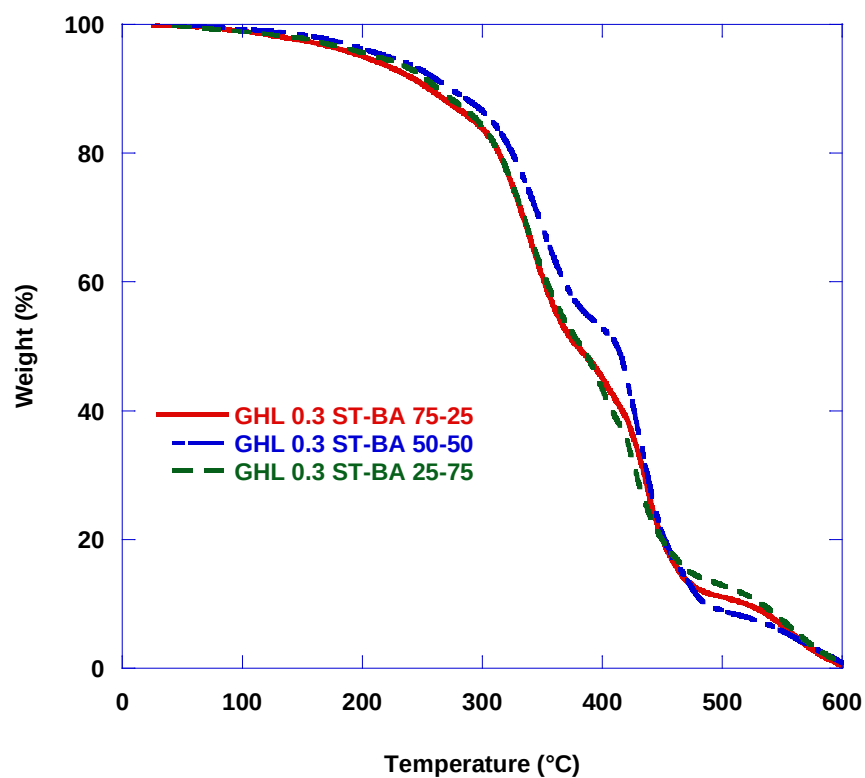


Figure 10: TGA curves for GHL films with different ratios of vinyl monomers (with HEA)

Table 6: Summary of TGA results

	HEA	ST	BA	Temp (°C) at 5% Weight Loss	Temp (°C) at 50% Weight Loss
GHL	0.0	25	75	231	383
		50	50	231	391
		75	25	237	388
	0.1	50	50	228	392
	0.2	50	50	228	403
	0.3	25	75	214	383
		50	50	224	413
		75	25	204	381
PU	0.0	-	-	232	373
	0.1	-	-	218	361
	0.2	-	-	220	359
	0.3	-	-	219	357

DMA DATA

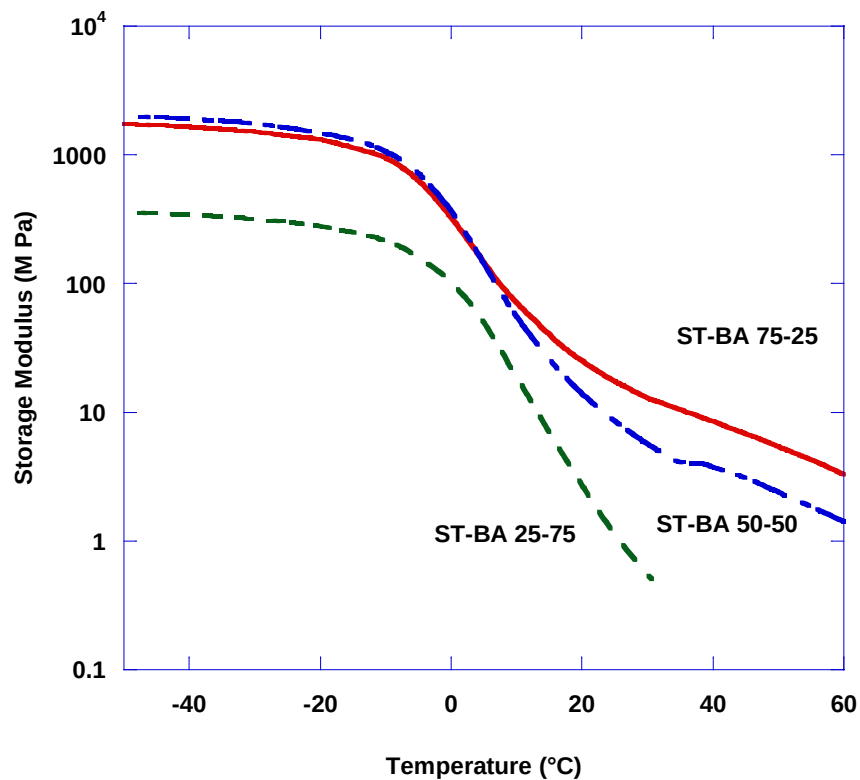


Figure 11: Storage modulus as a function of temperature for GHL films with different vinyl monomer ratios (without HEA)

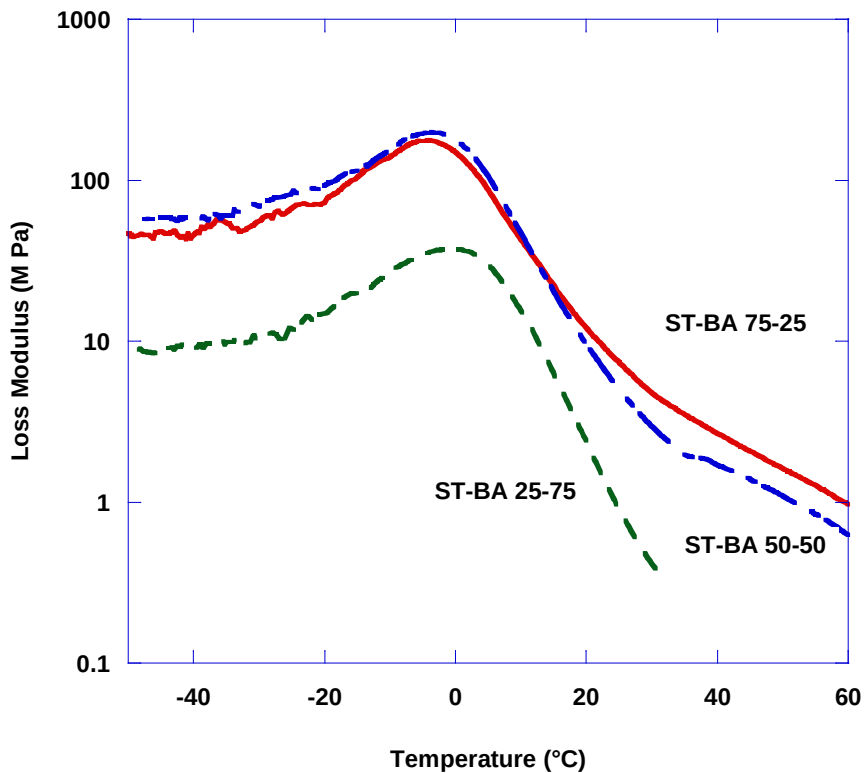


Figure 12: Loss modulus as a function of temperature for GHL films with different vinyl monomer ratios (without HEA)

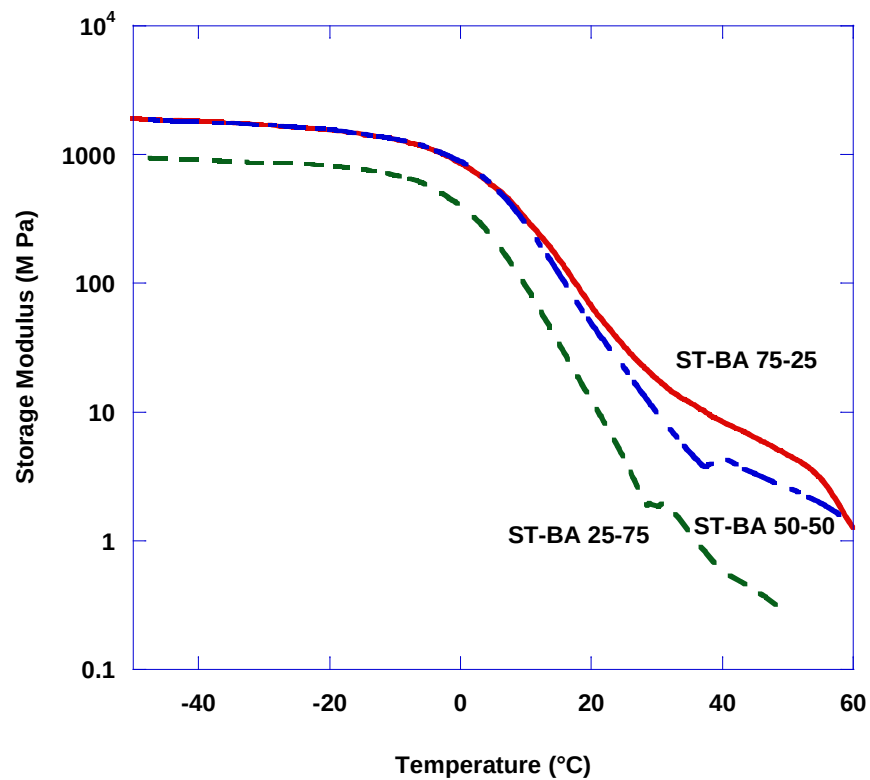


Figure 13: Storage modulus as a function of temperature for GHL films with different vinyl monomer ratios (with HEA)

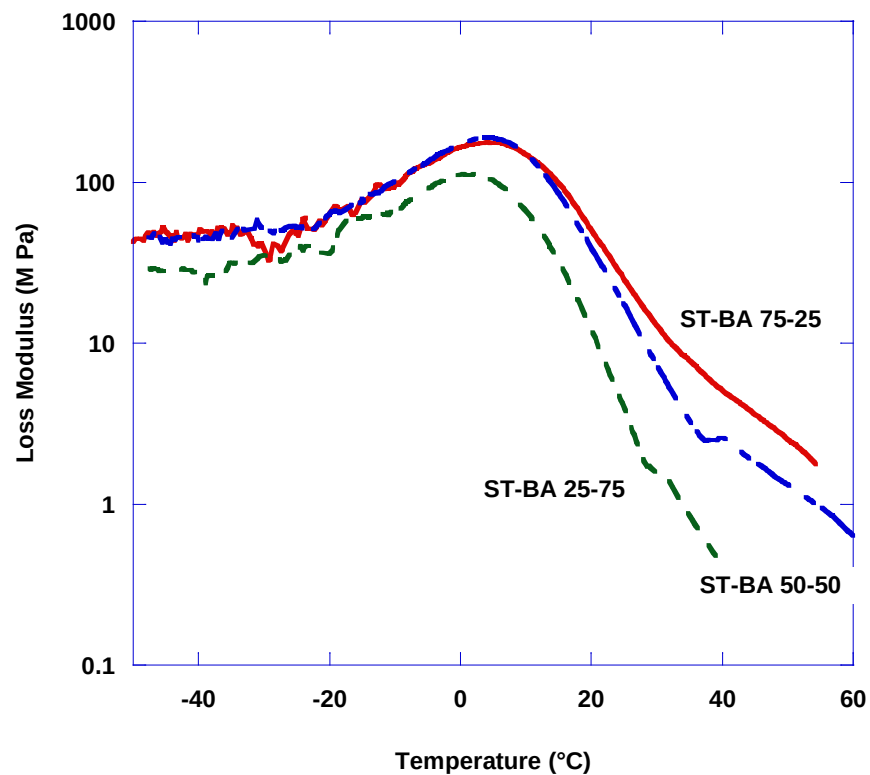


Figure 14: Loss modulus as a function of temperature for GHL films with different vinyl monomer ratios (with HEA)

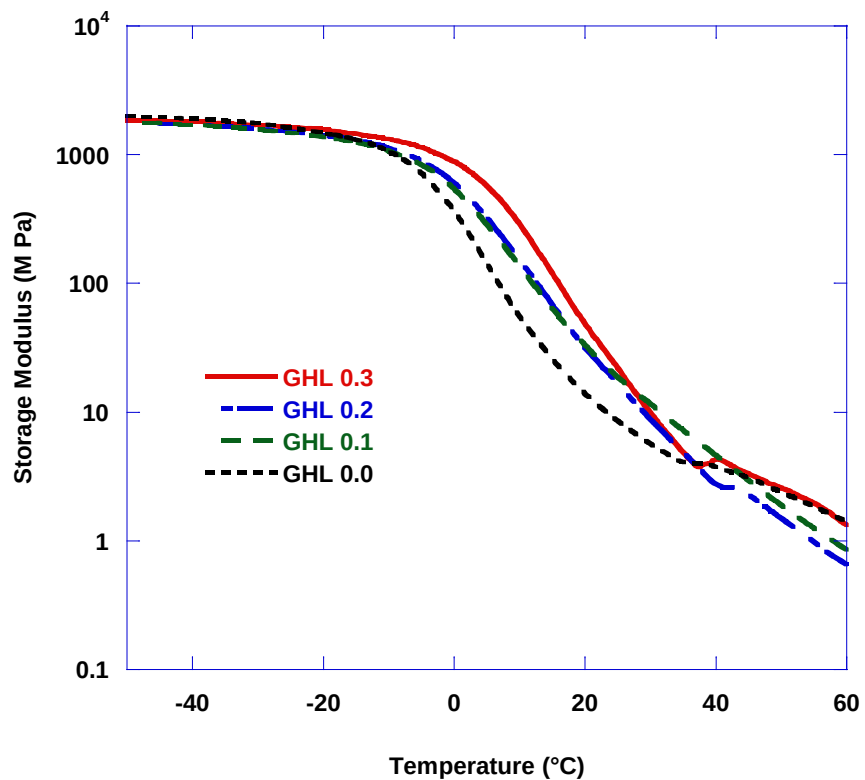


Figure 15: Storage modulus as a function of temperature for GHL films with different amounts of HEA

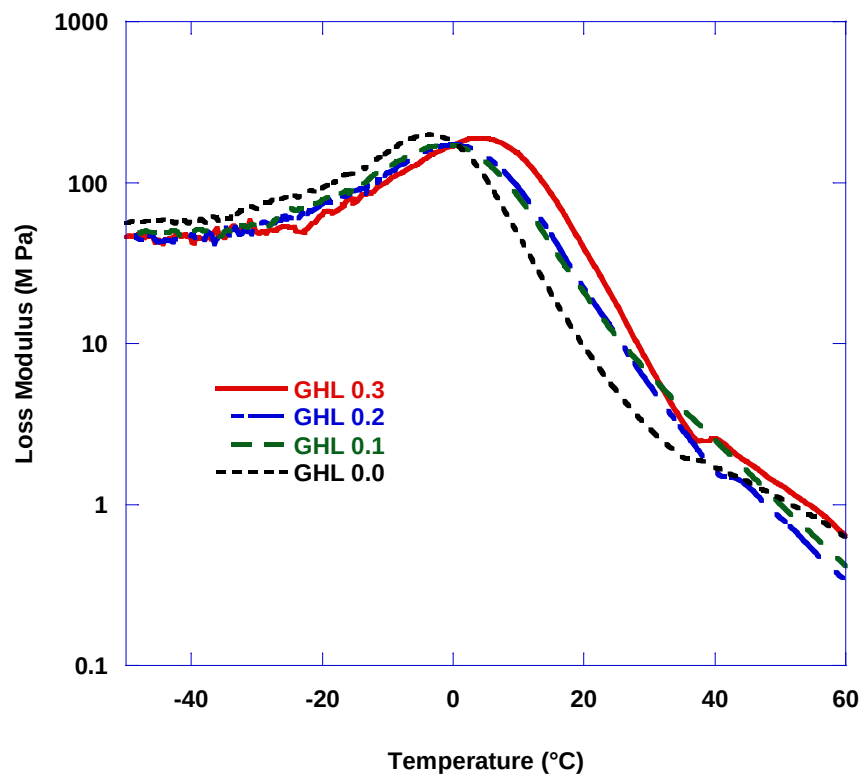


Figure 16: Loss modulus as a function of temperature for GHL films with different amounts of HEA

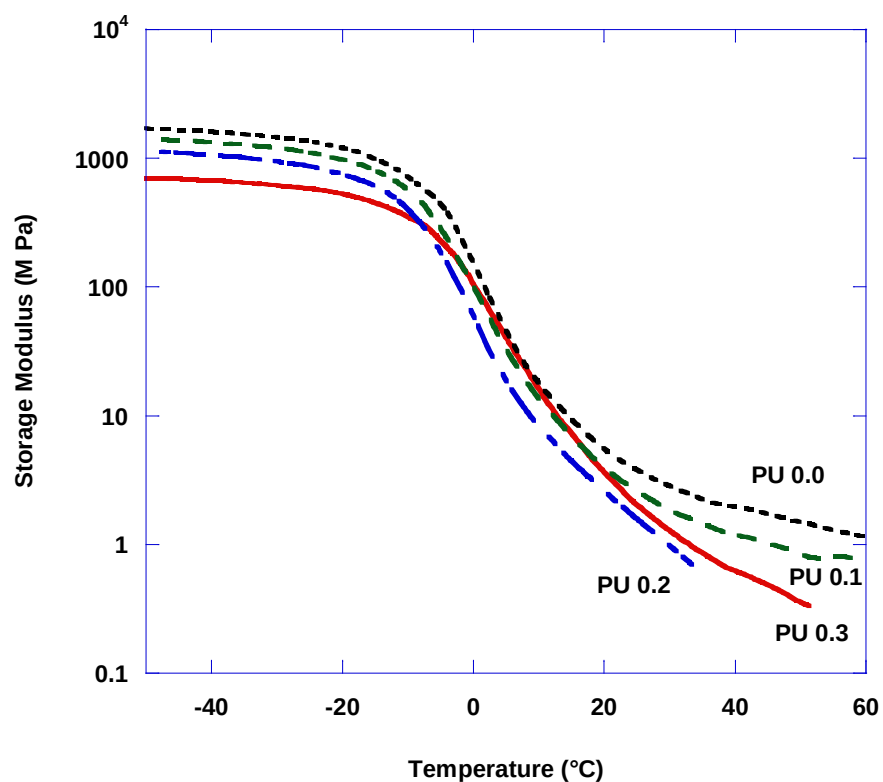


Figure 17: Storage modulus as a function of temperature for PU films with different amounts of HEA

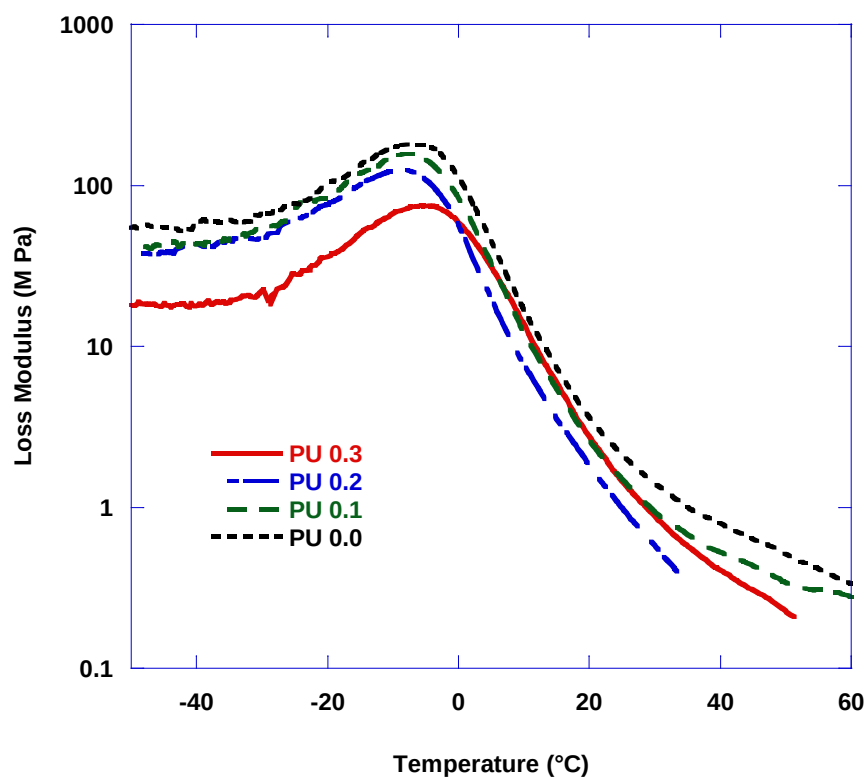


Figure 18: Loss modulus as a function of temperature for PU films with different amounts of HEA

TEM IMAGE

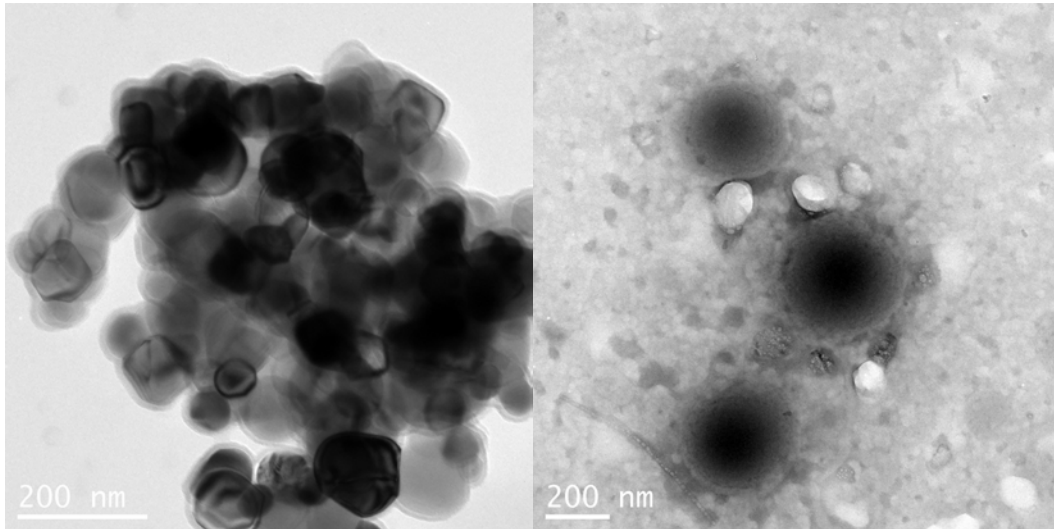


Figure 19: TEM images of GHL dispersion (0.2 HEA). The image on the left was unstained. The image on the right was stained with osmium tetroxide.

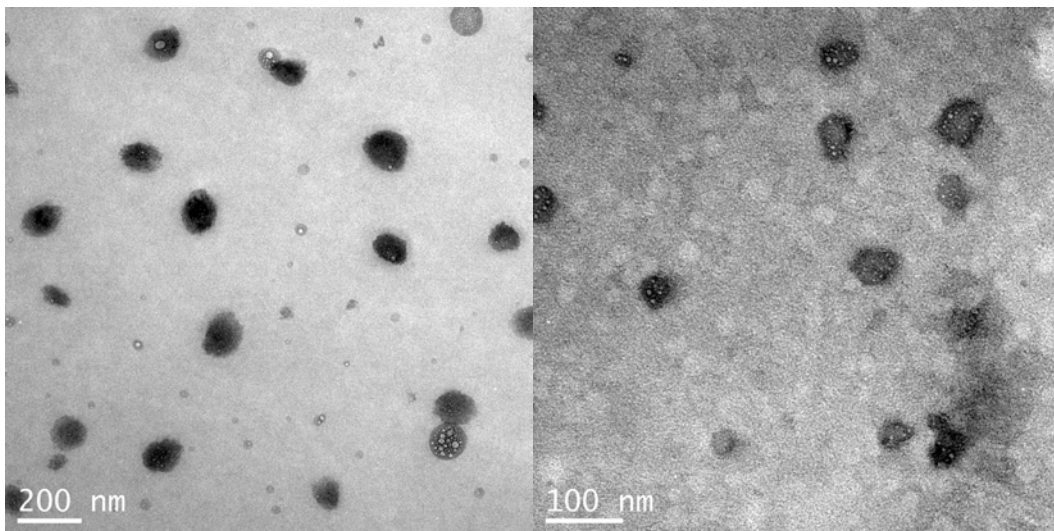


Figure 20: TEM images of GHL dispersion (0.3 HEA). The image on the left was unstained. The image on the right was stained with osmium tetroxide.

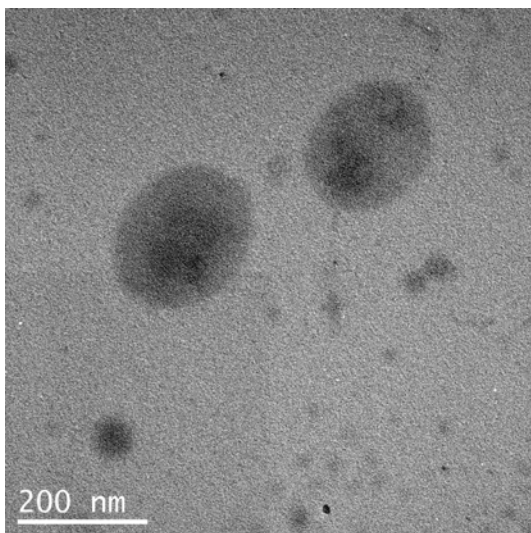


Figure 21: TEM image of PU dispersion (0.0 HEA) stained with osmium tetroxide.